

ORIGINAL RESEARCH

Farm-related injury deaths in Australia (2001–20)

Tony Lower PhD¹  | Kerri-Lynn Peachey BAS¹ | Margaret Rolfe PhD²¹AgHealth Australia, University of Sydney, Dubbo, NSW, Australia²University Centre for Rural Health, University of Sydney, Lismore, NSW, Australia

Correspondence

Tony Lower, AgHealth Australia, University of Sydney, PO BOX 256, Moree, NSW 2400, Australia.

Email: tony.lower@sydney.edu.au

Abstract

Objective: To describe the agents, pattern and trends of unintentional farm fatalities in Australia (2001–2020).**Methods:** Review of study of National Coronial Information System data.**Design:** Descriptive.**Setting:** Australia (2001–2020).**Participants:** All cases involving fatal work and non-work injury events on a farm.**Main outcome measures:** Patterns and agents of injury events, with trends for all-cause deaths based on rates per 10 000 farms and work-related incidents (per 100 000 workers and 1 000 000 h worked).**Results:** There were 1584 unintentional farm fatalities (annual mean 79). Two-thirds of cases were work-related (68%). Major agents of injury were farm vehicles (39%) and machinery (26%). Persons aged over 55 years were involved in 58% of all work-related incidents and were significantly more likely to die than younger cohorts when assessed against hours worked. Death rates involving all on-farm fatal incidents (both work and non-work) per 10 000 farms ($p = 0.015$) and work-related rates per 100 000 workers ($p = 0.015$) reduced over the period, with both demonstrating a fluctuating rate. There was no change in the work-related rates when assessed against hours worked ($p = 0.276$).**Conclusion:** Over the period, the annualised number of deaths fell by approximately 24% (98–75), with agents of injury remaining similar. General trends suggest a reduction in the overall death rates for work and non-work incidents. However, trends were less apparent when the reduction of farms (~19%), workers (~7%) and hours worked (no change), were accounted for. Targeted approaches are required to stimulate improvements in these preventable incidents.

KEYWORDS

ageing, agriculture, farm, injury

1 | INTRODUCTION

Internationally, agriculture is considered as a high-risk industry.^{1–3} In Australia, agriculture is one of the four key priority areas identified by the national work health and safety agency (Safe Work Australia) for specific interventions to reduce the injury burden.

The earliest Australian data to assess farm-related fatalities (1989–1992) identified that there was an annual mean of 146 unintentional farm deaths.⁴ More recent data from Safe Work Australia also clearly illustrate the disproportionate injury burden in the agricultural sector, with the all industries fatality rate being 1.5 per 100 000 workers, while the agricultural rate is some eight times higher at 12.7 per 100 000 over the same 2015–19 period.⁵

A previous assessment of fatal incidents in Australia assessing the 2001–15 period identified that while there had been a significant reduction in work-related death rates from the 1989–92 data,⁴ rates had remained virtually unchanged since 2001.⁶ Furthermore, around one-third of all farm deaths were identified as not being work-related. Importantly, official data from Safe Work Australia do not account for cases that are not deemed work-related. However, as the majority of Australian farms are family owned (>95%),⁷ the boundaries between the workplace, paid/unpaid work, family and home environments are frequently blurred. As such, it is important to consider all of these aspects to enhance health and safety for farmers, farm workers, family members and farm visitors.

Accurate data to underpin potential intervention points that effectively address the major farm injury risks are critical to reduce the injury burden associated with agricultural production. Consequently, this descriptive paper aims to describe the agents, pattern and trends of unintentional farm fatalities in Australia over the 2001–2020 period.

2 | METHODS

The National Coronial Information System (NCIS) is the central database of information about every death reported to an Australian coroner since July 2000 (January 2001 for Queensland) and to a New Zealand coroner from 1 July 2007.⁸ This study is restricted solely to Australian farm-related incidents, with all cases included where: (i) the person died unexpectedly and the cause of death is unknown; (ii) the person died in a violent or unnatural manner; and (iii) a doctor has been unable to sign a death certificate giving the cause of death. Farm fatalities are defined as those unintentional injuries occurring to farmers and people working in agricultural production, as well as to those in the farm workplace as bystanders to work

What is already known on this subject:

- Agriculture is a high-risk industry, where older farmers are known to be at increased risk of fatal injury
- Farm vehicles and machinery have consistently been identified as trauma sources

What this paper adds:

- Progress on reducing the fatal injury burden has stagnated over the past 20 years
- Rates of fatal farm injuries per million hours worked were significantly higher for those aged over 65 years
- There are significant variations in the rates of fatal farm injuries when examined by state, with South Australia the lowest and the Northern Territory the highest
- Highly focused and evidence-based interventions drawing on the hierarchy of risk controls are required to effectively address injury-related issues

being undertaken, and others in the course of leisure but harmed by hazards used in farm production. For each case, preliminary information is uploaded into the NCIS and these remain 'open' until the coroner hands down a final determination and the case is then 'closed'. In each of the NCIS cases, a cause of death is determined and recorded by a coroner, with specific cause of death details independently coded by the Australian Bureau of Statistics against the International Classification of Disease 10 (ICD-10).⁹

Identification of cases from NCIS for analyses involves two inter-related processes. First, a commercial media tracking organisation (iSentia) is used to scan approximately 2500 daily, weekly and monthly publications Australia wide. Publications are scanned for various designated search terms (e.g. 'farm*', 'agric*', 'growers', 'producers' and 'horticulture'). Where a potential on-farm case is identified, the corresponding NCIS case file number is obtained for this 'open' case. However, there is potential for cases to have been missed using this strategy, as not all cases are reported in the media. Consequently, the second approach to identifying cases of relevance relies on keyword searches of the NCIS (farm* OR agric*) for each year. These are then reviewed with cases that are not farm-related and those confirmed as intentional by the Coroner, being withdrawn from the dataset.

The available NCIS data for farm cases are coded using the Farm Injury Optimal Dataset, with fatalities including both work and non-work-related activities.¹⁰ Each case is

coded in relation to demographics, role in event (e.g. operator, bystander, and passenger), work relatedness, relevant causal agents of injury (dams, tractors, grain augers, quads etc.), mechanism of injury, toxicological results (alcohol/drugs) and other context-specific information as applicable, such as helmet usage, loads and rollover/runover. This coding format has been widely used in other Australian farm injury studies.^{11–13}

Descriptive statistics in relation to age, gender, causal agents and mechanisms of injury were calculated. Three year rolling averages for case numbers were used over the 20-year period. Denominator data on the number of persons working in the agricultural sector, hours worked and the number of farms nationally and by state, were derived from the Australian Bureau of Statistics and the Australian Bureau of Agricultural and Resource Economics and Sciences.^{14,15} Data on hours worked for those aged 15+ years (million hours) by state and age group were collected in the Census data (2001, 2006, 2011, and 2016) and interpolated in the intervening years. Where working hours were reported as a range, for example 16–24 h per week, the mean number of hours (i.e. 20) was applied to calculate total hours worked. For the 2016 to 2020 period, the data for the 2016 Census were used as a constant.

From 2016, there was a definitional change by the ABS in determining farm numbers. Up to 2015, farms with an estimated value of agricultural output (EVAO) of >\$5000 were included in the data; however, this was altered to >\$40000 in 2016. To account for this change, data were derived from the ABS indicating that of the total number of farms in 2015 (EVAO >\$5000), 31.2% had an EVAO between \$5000 and \$40000. This ratio was applied to the post 2016 data to attain comparative estimates of all farms with an EVAO >\$5000. In addition, as farm number data for 2020 were as yet unavailable, the number of farms calculated for 2019 was used as a proxy.

Crude rates were determined for all deaths (work and non-work) per 10 000 agricultural establishments. Rates were also determined for all work-related deaths per 100 000 workers for persons older than 15 years by age cohort and per 1 000 000 h worked, both nationally and by state. Comparisons between age cohorts and states were assessed using incidence rate ratios, with 95% confidence intervals. Reference groups were those with the lowest incidence rate.

Due to the count nature of these data, polynomial (linear, quadratic and Cubic) Poisson regression analyses with numbers of deaths and offsets as the respective target populations presented earlier, were completed to assess the temporal pattern over the 2001–2020 period.¹⁶ For modelling, year was centred at 2010, thus 2010 became 0, 2020 10, 2000 became –10. Change and model

improvement was considered statistically significant at $p < 0.05$. IBM SPSS (Version 26) was used for these analyses. The study has ethics approval from the Department of Justice Human Research Ethics Committee—Approval number CF/19/27527.

3 | RESULTS

In the period 2001–2020, there were 1584 unintentional farm fatalities with 1412 (89%) finalised and closed in the NCIS and 172 (11%) open (Table 1). Over the period, the total number of annual deaths dropped from 98 to 75 (24%). The annual mean number of deaths was 79 (SD 12.2), consisting of 54 (SD 8.7) work cases and 25 (SD 9.7) non-work cases. Work-related cases dropped from approximately 66 to 53 per annum (~20%), while non-work incidents fell from 31 to 18 per annum (~42%). Men accounted for 1398 (88%) of decedents.

Rates of death involving all on-farm fatal incidents (both work and non-work) per 10 000 agricultural establishments demonstrated significant effects over time. The linear model ($b = -0.009$, $SE = 0.0044$, $p = 0.035$) demonstrated a statistically significant decline in the death rate. However, the quadratic model (linear term $b = -0.011$, $SE = 0.0043$, $p = 0.014$; quadratic term $b = 0.002$, $SE = 0.0008$, $p = 0.015$) was a significant improvement to the linear model ($p = 0.016$). The addition of a cubic term was not a significant improvement to the model ($p = 0.119$). The mean death rate was 5.9 per 10 000 farms (Range 5.0–7.4; SE 0.17), moving from 7.3 in 2001 to 5.6 in 2020.

Work-related rates (per 100 000 workers) revealed a significant cubic relationship over time (Omnibus test $p = 0.075$), with significant linear ($b = 0.033$, $SE = 0.0136$, $p = 0.014$) and cubic terms ($b = -0.0005$, $SE = 0.0002$, $p = 0.015$) and non-significant quadratic term ($b = 0.001$, $SE = 0.0020$, $p = 0.189$). The linear and quadratic models (linear term only; linear + quadratic) were non-significant ($p = 0.498$ and $p = 0.451$, respectively). This demonstrated a curvilinear response and the fluctuating nature of this rate. Overall, the mean work-related mortality rate (per 100 000 workers) was 17.6 (Range 12.6–21.4; SE 0.56), moving from 18.5 in 2001 to 16.5 in 2020.

When examined in relation to million hours worked, none of the polynomials fitted (Omnibus test linear $p = 0.276$; quadratic $p = 0.428$; cubic $p = 0.143$) were significant, thus a constant rate over time was demonstrated. The mean rate over the period was 0.12 (Range 0.08–0.15; SE 0.003).

Figure 1 illustrates the pattern over time for million hours worked (including 95% confidence limits). Similar curvilinear patterns were also identified for work-related

TABLE 1 Case numbers by year, work-relatedness, rates per 10 000 farms, rates per 100 000 workers and rates per million hours worked.

Year	No. cases	Rolling average (3 Yr)	Work-related (total)	Work-related (15+ total)	Non-work	Unknown	No. farms	Rate/no. farms (10000)	No. workers	Rate/workers (100 000)	Hours worked	Rate/million hours worked
2001	102		70	69	32		140 516	7.3	372 900	18.5	529 048 259	0.13
2002	92		63	63	29		137 377	6.7	351 325	17.9	509 766 652	0.12
2003	99	98	66	64	33		132 983	7.4	317 775	20.1	490 485 045	0.13
2004	95	95	65	63	29	^a	130 526	7.3	308 575	20.3	471 203 438	0.13
2005	70	88	38	38	32		129 934	5.4	302 375	12.6	451 921 831	0.08
2006	78	81	39	39	39		154 680	5.0	299 200	13.0	432 640 222	0.09
2007	83	77	47	46	36		150 817	5.5	300 450	15.3	424 147 014	0.11
2008	71	77	50	49	21		140 704	5.0	297 375	16.5	415 653 806	0.12
2009	75	76	53	51	22		135 996	5.5	307 725	16.6	407 160 598	0.13
2010	72	73	48	47	24		134 553	5.4	310 375	15.1	398 667 390	0.12
2011	82	76	54	52	28		135 563	6.0	273 025	19.0	390 174 180	0.13
2012	75	76	55	54	20		135 692	5.5	276 300	19.5	392 213 827	0.14
2013	89	82	55	53	34		128 917	6.9	259 250	20.4	394 253 474	0.13
2014	68	77	46	45	22		128 489	5.3	279 950	16.1	396 293 121	0.11
2015	70	76	59	58	11		123 091	5.7	271 200	21.4	398 332 768	0.15
2016	66	68	49	45	17		124 536	5.3	266 500	16.9	400 372 416	0.11
2017	72	69	56	55	16		126 337	5.7	263 850	20.8	400 372 416	0.14
2018	79	72	57	55	22		128 140	6.2	283 950	19.4	400 372 416	0.14
2019	73	75	50	48	17	6	129 942	5.6	283 900	16.9	400 372 416	0.12
2020	73	75	51	51	16	6	129 942	5.6	309 400	16.5	400 372 416	0.13
Total	1584		1071	1042	500	13	2 678 735	118	5 935 400	352	8 503 823 707	2.46
Mean	79.20		53.55	52.10	25.00		133 637	5.9	296 770	17.6	425 191 185	0.12

^a >5 cases.

rates per 100 000 workers and rates of death per 10 000 farms (work and non-work). In each instance, there was a reduction in the incidence of cases from 2001 to approximately 2006–07, followed by an increase up to 2016 and another slight reduction through to 2020.

3.1 | Age and work-related deaths

In relation to the age of persons fatality injured while working (Table 2), the lowest incidence of cases was in persons aged 35–44 years (0.07). Meanwhile, the number of cases involving persons over 65 years exceeded one-third of all cases ($n = 374$). The 65+ (IRR = 4.63) age cohort had a significantly higher incidence rate ratio when compared with all other ages.

3.2 | Deaths by state

South Australia had a statistically significant lower rate of injury than all other states (Table 3). In contrast, data for the Northern Territory illustrated the highest rate (0.38) and incidence rate ratio (5.52), with Tasmania also having an elevated rate (0.26) and incidence rate ratio (3.73).

3.3 | Agents of fatal injury

Farm vehicles were involved in 39% ($n = 618$) of cases, mobile farm machinery 25.8% ($n = 409$), farm structures 11.4% ($n = 181$), animals 8.2% ($n = 130$) and working environment 7.3% ($n = 116$). More specifically, the leading agents as outlined in Table 4 were as follows: tractors (15.8%); quads (all-terrain vehicles—ATVs) (12.3%); water

bodies (6.3%; $n = 100$ —including dams 3.8%; $n = 60$); farm utilities (6.6%); motorbikes (5.2%); horses (4.7%) and tree felling (3.4%). In sum, these factors alone accounted for almost 58% of all fatal cases.

4 | DISCUSSION

Data from the NCIS capturing all non-intentional fatal injuries on Australian farms in the 2001–20 period indicated small but statistically significant reductions for all deaths (work and non-work) when assessed per 10 000 farms and for work-related deaths per 100 000 workers. However, work-related deaths per million hours worked (mean 0.12), illustrated no significant changes over the 20-year time frame. The overall pattern of injury-related deaths was curvilinear in nature, with reductions through to 2006–07, followed by an increase up to 2016 and another slight reduction to 2020. These data are consistent with previous findings assessing the earlier 2001–15 period.⁶

Almost 70% of cases were work-related, with men being disproportionately represented in incidents (88%). Persons aged 65+ (0.32 per million hours worked) had significantly higher rates of work-related death than all other age groups, with those aged 55–64 years (0.13) also being significantly more likely to be involved in these incidents than the reference group (35–44 years). These findings reflect similar patterns in other western agricultural systems and within Australia.^{17–19} Persons working in Tasmania and the Northern Territory were also at increased risk in comparison with other jurisdictions, which is also in line with earlier data.⁶

Major agents of injury accounting for over two-thirds of incidents have not altered markedly over the period

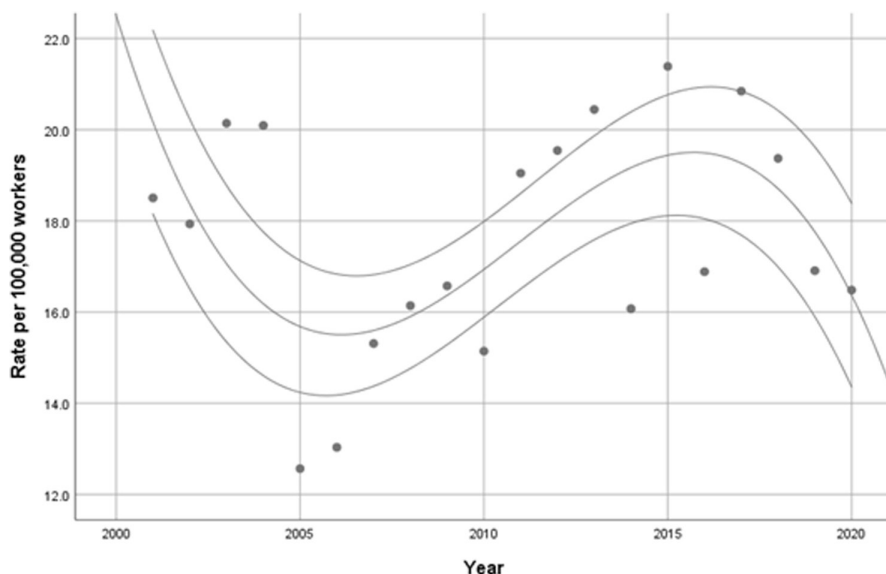


FIGURE 1 Cubic model of work-related deaths (15+ years) per million hours worked from 2001 to 2020 with 95% confidence limits.

TABLE 2 Incidence rate ratio for age cohorts based on million hours worked in the 2001–2020 period.

Age	No. cases	%	Hours worked	Rate ^a	Rate ratio	95%CI
35–44	113	10.9	1 637 231 209	0.07		
15–24	79	7.5	681 665 719	0.11	1.66	0.80–3.42
25–34	88	8.5	1 221 157 683	0.07	1.04	0.78–1.37
45–54	157	15.1	1 993 354 957	0.08	1.14	0.89–1.45
55–64	230	22.1	1 800 243 627	0.13	1.85	1.47–2.31
65+	375	36.0	1 169 533 453	0.32	4.63	3.75–5.70
Total	1042		8 503 186 667	0.12		

^aRounded to two decimal places.**TABLE 3** Incident rate ratio for Australian states/territories based on million hours worked in the 2001–2020 period.

State	No. cases	Hours worked	Rate	Rate ratio	95%CI
SA	61	891 477 256	0.07		
WA	88	851 542 970	0.10	1.51	1.09–2.09
VIC	232	1 863 982 582	0.12	1.82	1.37–2.41
NSW	309	2 314 811 075	0.13	1.95	1.48–2.56
QLD	270	1 886 881 271	0.14	2.09	1.58–2.76
TAS	61	239 088 179	0.26	3.73	2.05–4.61
NT	23	609 110 38	0.38	5.52	3.44–8.83
Total	1042	8 193 743 793	0.19		

and centred on farm vehicles (39.0%) and mobile farm machinery (25.8%). Smaller burdens were associated with farm structures (11.4%), animals (8.2%) and the working environment (7.3%).

Whilst these data are suggestive of little change in work-related cases, there does appear to have been roughly a numeric halving of cases on an annual basis that are not involving work. Such positive progress is important to recognise; however, there are considerations in relation to the metric used (rate per 10 000 farms), that need clarification. First, the metric uses all cases (inclusive of work and non-work) as the numerator. As Australian farms are predominantly family-owned businesses and frequently serve as both a residence and a workplace, this increases potential risks to those in the environment, including family members and visitors. Second, the overall number of farms used as a denominator has dropped by around 11 000 due to the aggregation of smaller farms into larger holdings over the 20-year period. Third, rural demographics in Australia indicate populations (including those living on farms), are both ageing and notwithstanding recent internal migration as a result of COVID-19, decreasing in number.²⁰ Consequently, these factors impact on potential exposure to hazards and necessitate that due caution should be applied in interpreting these data.

For the work-related cases, the evidence appears more robust given the nature of the denominator data. While a statistically significant reduction in rates per 100 000 workers is welcomed no improvement was identified when the rate per 1 000 000 h worked was assessed, although a similar curvilinear pattern was identified. There is no definitive rationale underpinning the reduction through to 2006–07 and subsequent increments and reductions. Although overall annual case numbers (mean 79) are relatively small in a statistical sense and could be expected to fluctuate over time, the increment does coincide with the cessation of government funding to Farmsafe Australia supporting the conduct of preventive programs. While this situation has recently been rectified,²¹ further assessment is required to substantiate impacts of such broad-based programs. While it is positive that rates have not increased substantially, it is equally disappointing that minimal progress has been made to further drive these rates down. This is particularly relevant when comparisons are made of the agricultural data in this study (17.6 per 100 000), with the all-industries rate in Australia (1.5 per 100 000 workers), a differential that is almost 12 times higher.⁵

The average age of Australian farmers has now exceeded 58 years,²² with ageing identified as a risk factor in several previous assessments internationally.^{18,23} One

TABLE 4 Agents involved in all unintentional farm deaths (work and non-work related) in Australia, 2001–2020.

Agent	n	%	Agent	n	%
Farm vehicles	618	39.0	Mobile farm machinery	409	25.8
Quad (ATV)	195	12.3	Tractor	251	15.8
Utility (pick-up)	105	6.6	Forklift	16	1.0
Motorbike	83	5.2	Bulldozer	14	0.9
Car	50	3.2	Harvester	13	0.8
Fixed wing plane	40	2.5	Slasher	15	0.9
Helicopter	33	2.1	Grain auger	8	0.5
Truck	38	2.4	Cherry picker	7	0.4
Trailer	16	1.0	Excavator	6	0.4
Side by Side Vehicle	37	2.3	Power take off	6	0.4
Gyrocopter	9	0.6	FEL	6	0.4
Ultralight	5	0.3	Other	67	4.2
Other	7	0.4			
Farm structures	181	11.4	Working environment	116	7.3
Dam	60	3.8	Trees (felling)	54	3.4
Power lines	24	1.5	Tree branch (not felling)	35	2.2
Other (water)	22	1.4	Other	27	1.7
Creek/river	9	0.6			
Pool	9	0.6	Animals	130	8.2
Shed	11	0.7	Horse	75	4.7
Silo	8	0.5	Cattle	31	2.0
Windmill	6	0.4	Other	24	1.5
Other	32	2.0			
Other Classified	124	7.8	Unknown	6	0.4
Firearms (non-intentional)	30	1.9			
Ride on mower	7	0.4			
Hay Bale	10	0.6			
Other	77	4.9			
			Total	1584	100

possibility underpinning the difficulty in further reducing rates in Australia could be the lack of direct engagement with this cohort of older farmers, with those over 55 years representing 58% of all fatal cases in these data. Any approach targeting this age cohort should ensure that it is based on the hierarchy of risk controls, be evidence-based and also aligned with contextual approaches that engage this cohort in enhancing safety.

The geographic distribution of fatal cases by state/territory, reinforced previous data demonstrating that the Northern Territory and Tasmania had significantly elevated rates of death.^{4,6} The Northern Territory is a vast area that is the least densely populated Australian state (0.16/km²),²⁴ with the dominate agricultural industry being broadacre beef cattle production. Whilst speculative in nature, possible contributory factors to the increased risk of fatality may be the nature of the work

itself, plus the major impediments to operationalising and enforcing health and safety procedures in such a large area. In contrast, Tasmania is the smallest state in Australia and has the fourth highest population density (7.24/km²).²⁵ It has a more broadly based enterprise mix consisting of sheep/wool, horticulture, beef and dairy cattle businesses; hence, what underpins this elevated risk is not clear.

A study limitation is that 11% of all cases remained ‘open’ within the NCIS and hence further detail will come to hand in relation to these cases over time. However, this is a relatively small component of the overall sample and what impact they may have on the final interpretation of the data are unknown. A significant strength of this assessment is that three potential indicators were utilised in conjunction (rates per 10 000 farms; per 100 000 workers; and, per 1 000 000 h worked), providing further validation

for the results. Of specific note are the data focusing on hours worked, as this provides an enhanced measure of risk exposure.

Overall, the data illustrate a small reduction in the profile for farm fatalities in the past 20-year period. While agricultural outputs continue to grow in Australia, there is a need to reinvigorate steps to reduce farm deaths and injuries. As per previous assessments, there is a disproportionate burden associated with farm vehicles and machinery that requires addressing. Similarly, the representation of older persons (55+ years) in the data, in conjunction with an ageing farmer population, provides a clear focus for attention. Failure to directly address these issues will result in little to no improvement in the health and safety record for the agricultural sector, which has significant impacts on individuals, families, businesses and whole communities.

AUTHOR CONTRIBUTIONS

TL: conceptualization; formal analysis; methodology; writing – original draft; writing – review and editing. K-LP: data curation; project administration; writing – review and editing. MR: formal analysis; writing – review and editing.

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ETHICS APPROVAL

All procedures performed in this study were in accordance with the ethical standards of the Department of Justice Human Research Ethics Committee—Approval number CF/19/27527 and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

ORCID

Tony Lower  <https://orcid.org/0000-0003-2243-6764>

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